

EC2 Website + Database Integration Setup

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Project: EC2 Website+ Database Integration setup

Objective

- Login AWS
- Project Purpose

Architecture Structure

- Structure
- Explanation

AWS Services Used

- Amazon EC2
- Amazon RDS
- Security Group
- Amazon VPC

Testing Outcome

- Created Successful
- Uploaded Successful

Conclusion

- Sunmmary
- Learning

1. OBJECTIVE-

- Login to AWS Console.
- Search RDS.
- Open the RDS Dashboard.

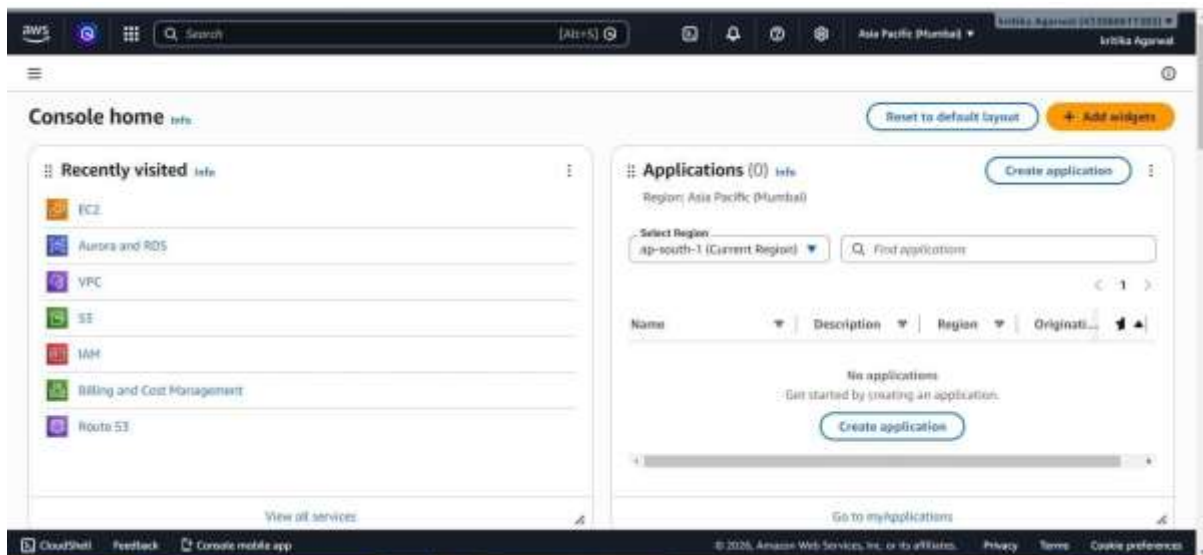


FIG-1: AWS CONSOLE HOME PAGE

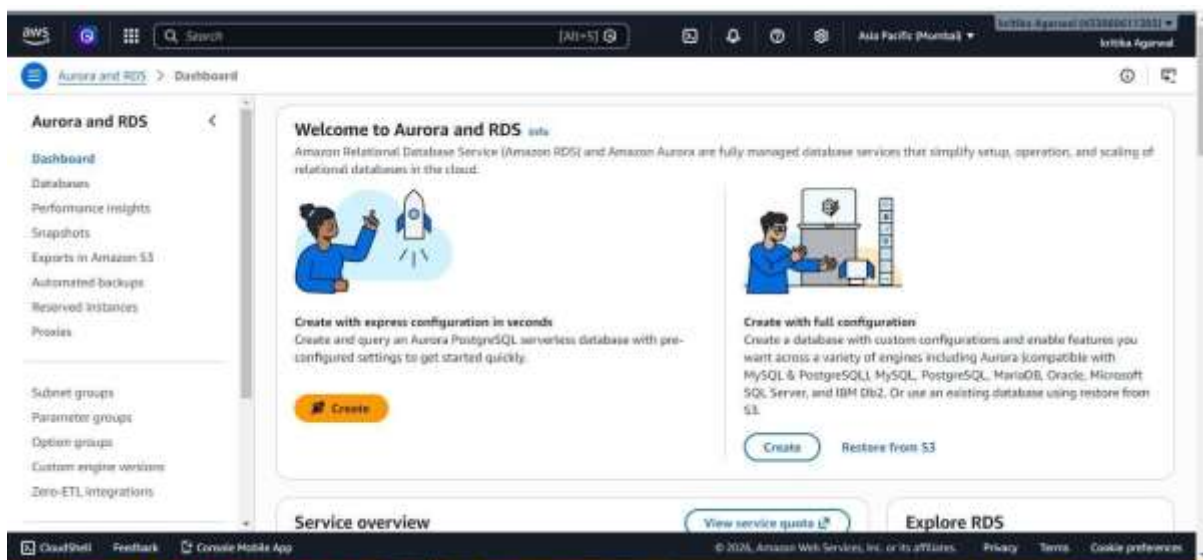


FIG-2: RDS DASHBOARD

2. ARCHITECTURE STRUCTURE-

- Users EC2 Public IP ke through website access karte hain.
- Amazon EC2 par Apache Web Server aur PHP application host ki gayi hai.
- PHP application MySQL (Port 3306) ke through Amazon RDS database se connect hoti hai.
- RDS database data ko securely store, read aur write karta hai.

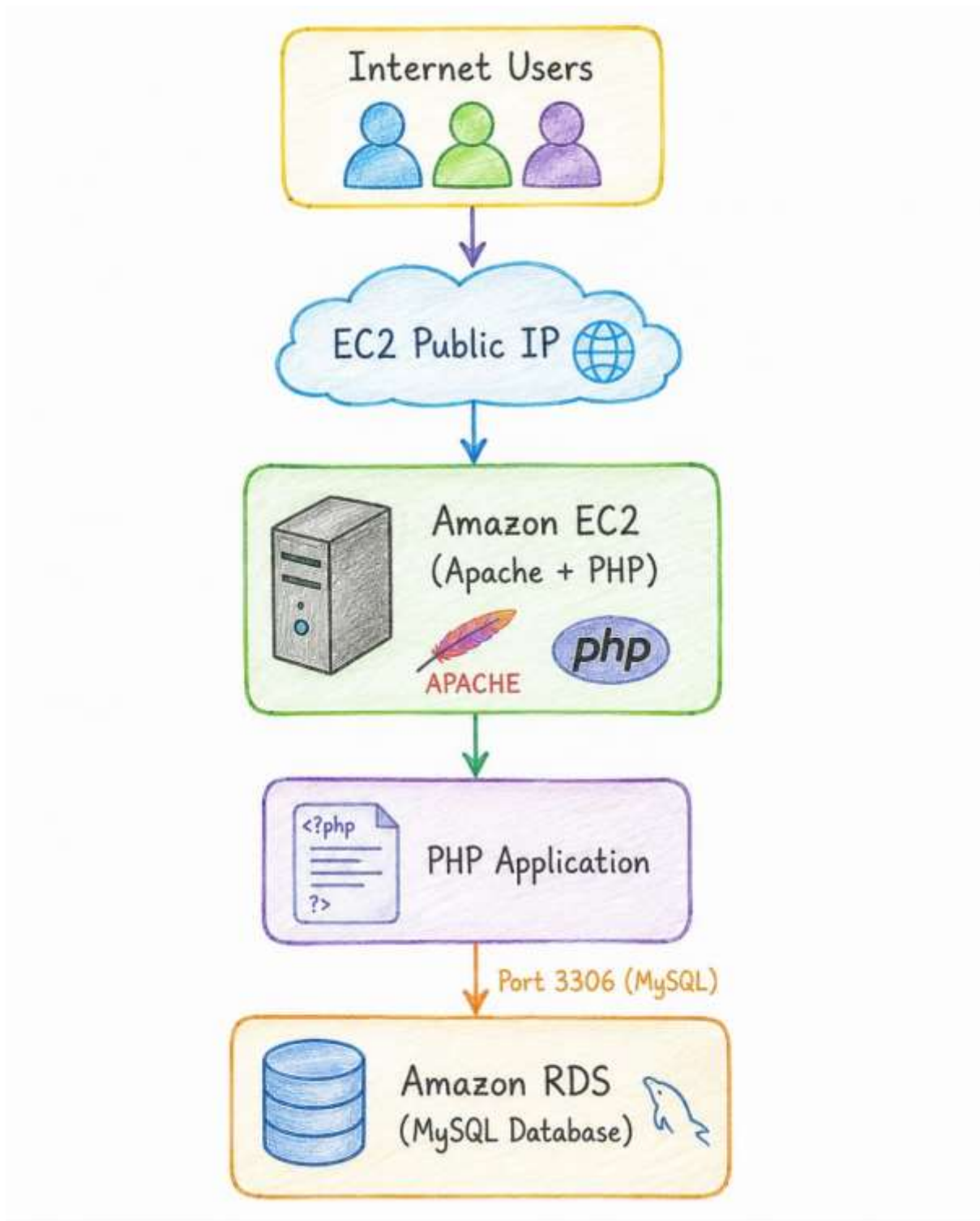


FIG: RDS ARCHITECTURE

3. AWS SERVICES USED-

➤ STEP 1- Launch EC2 Instance:

❖ Configuration-

- Name: WebServer

- AMI: Amazon Linux 2023
- Instance Type: t3.micro
- Key Pair: Select existing key
- VPC: Your project VPC
- Public Subnet: Enable
- Auto Assign Public IP: Enable

❖ Security Group-

Type	Port	Source
SSH	22	My IP
HTTP	80	Anywhere(0.0.0.0/0)
HTTPS	443	Anywhere(0.0.0.0/0)

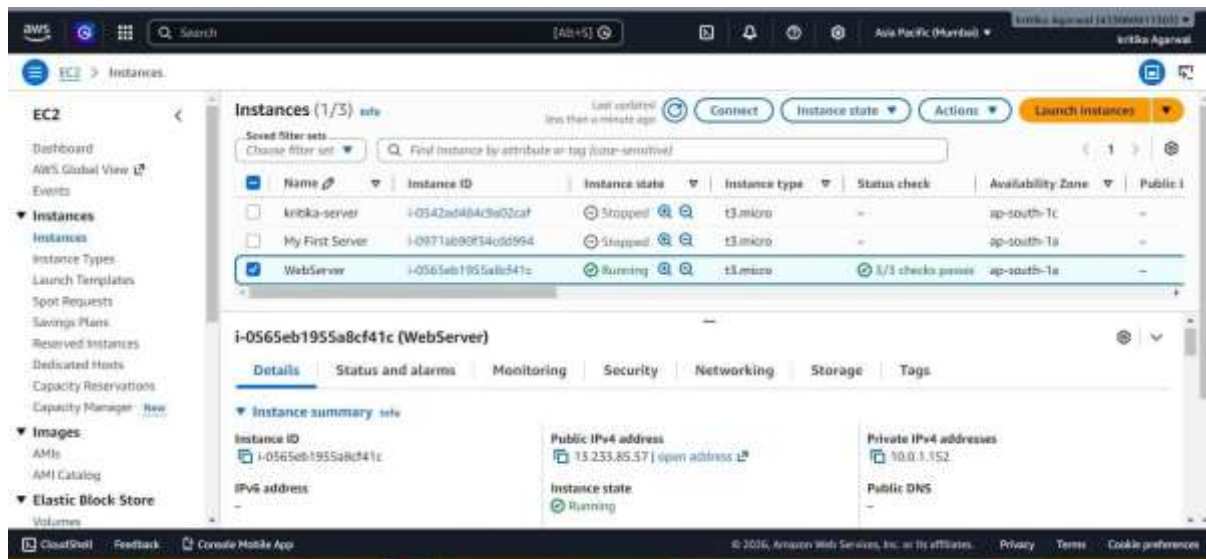


FIG-1: EC2 INSTANCE

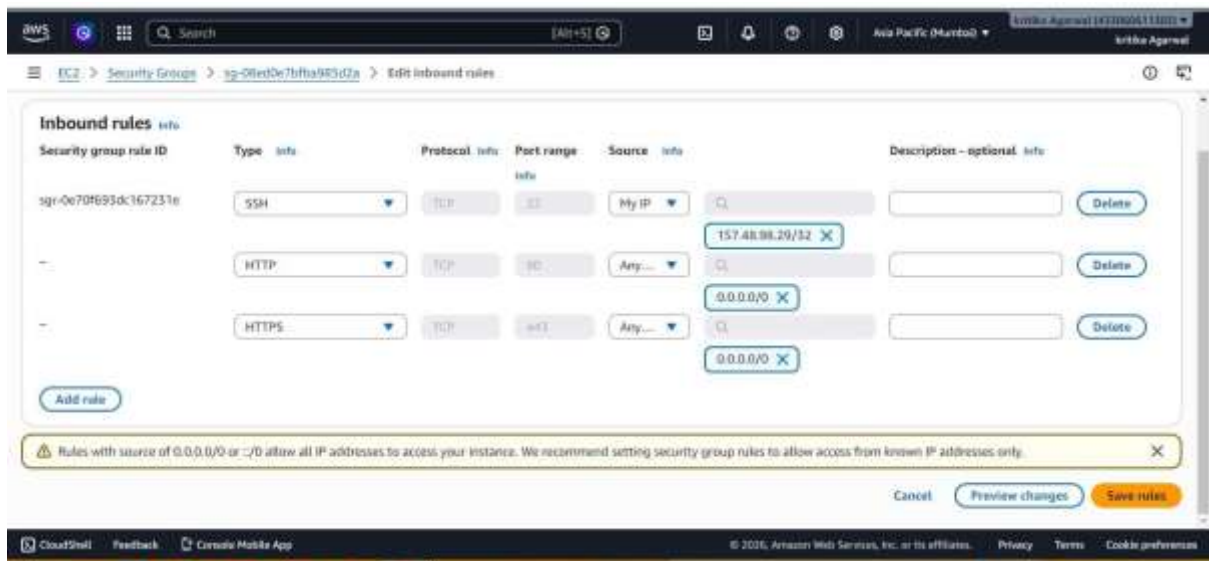


FIG-2: SECURITY GROUP

➤ STEP 2- Connect to EC2:

- Open PowerShell (Windows) or Terminal (Linux).
- `ssh -i your-key.pem ec2-user@<EC2-Public-IP>`

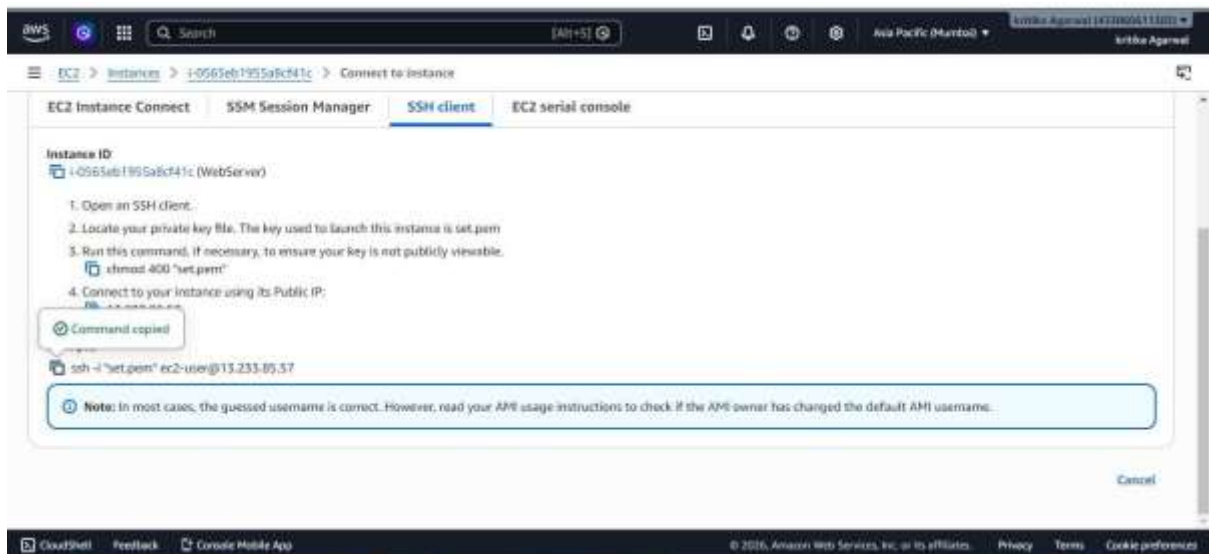
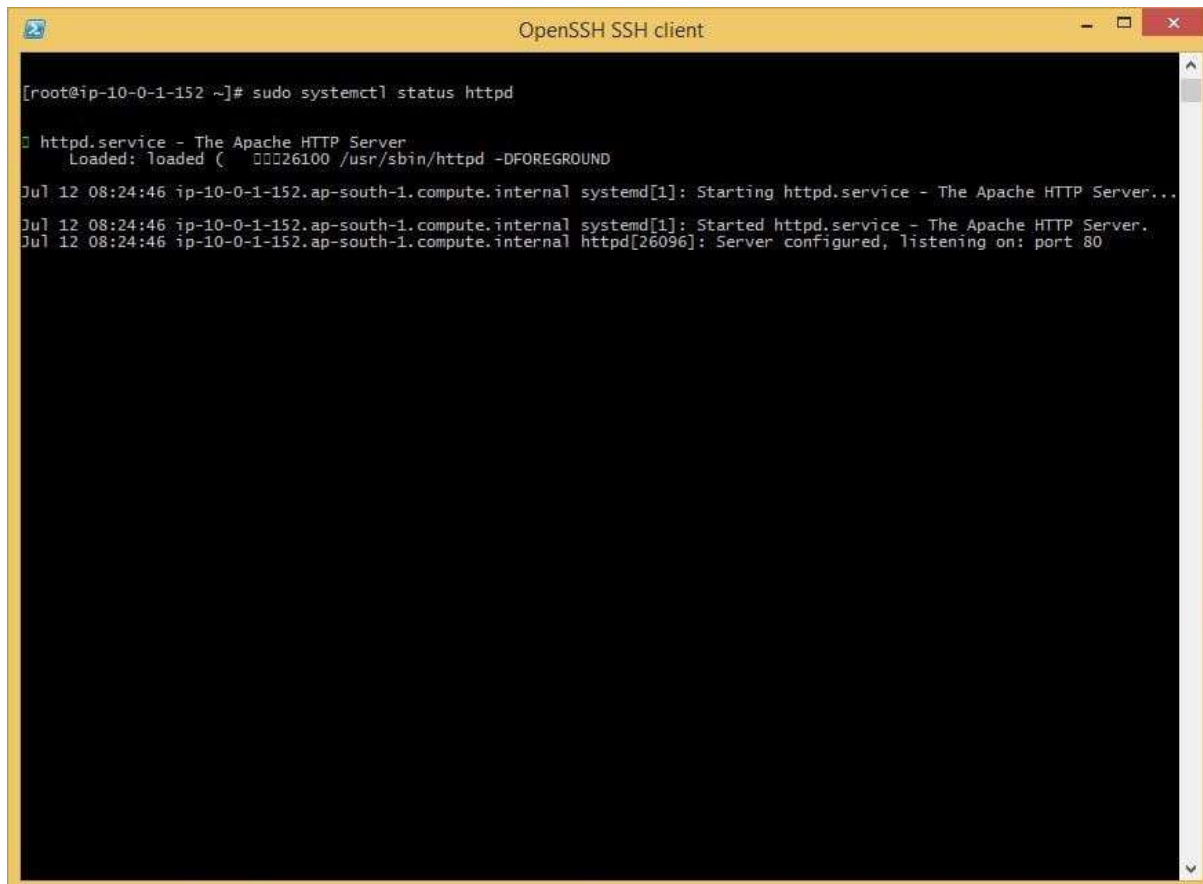


FIG-1: COMMAND PAGE


```
OpenSSH SSH client
e-user@ip-10-0-1-152 ~]$ sudo -i
[root@ip-10-0-1-152 ~]# sudo yum install httpd -y
Last metadata expiration check: 0:01:08 ago on Sun Jul 12 08:23:10 2026.
Dependencies resolved.
=====
Package                                Architecture      Version            Repository          Size
-----
Installing:
httpd                                  x86_64            2.4.68-1.amzn2023.0.1  amazonlinux         46 k
Installing dependencies:
apr                                    x86_64            1.7.5-1.amzn2023.0.4  amazonlinux         129 k
apr-util                              x86_64            1.6.3-1.amzn2023.0.2  amazonlinux          97 k
apr-util-ldap                         x86_64            1.6.3-1.amzn2023.0.2  amazonlinux          13 k
generic-logos-httpd                  noarch            18.0.0-12.amzn2023.0.3  amazonlinux          19 k
httpd-core                           x86_64            2.4.68-1.amzn2023.0.1  amazonlinux         1.4 M
httpd-filesystem                     noarch            2.4.68-1.amzn2023.0.1  amazonlinux          12 k
httpd-tools                          x86_64            2.4.68-1.amzn2023.0.1  amazonlinux          80 k
libbrotli                             x86_64            1.0.9-4.amzn2023.0.2  amazonlinux         315 k
mailcap                               noarch            2.1.49-3.amzn2023.0.3  amazonlinux          33 k
Installing weak dependencies:
apr-util-openssl                     x86_64            1.6.3-1.amzn2023.0.2  amazonlinux          15 k
mod_http2                            x86_64            2.0.42-1.amzn2023.0.1  amazonlinux         167 k
mod_lua                              x86_64            2.4.68-1.amzn2023.0.1  amazonlinux          59 k
=====
Transaction Summary
=====
Install 13 Packages
Total download size: 2.4 M
Installed size: 7.0 M
Downloading Packages:
(1/13): apr-1.7.5-1.amzn2023.0.4.x86_64. 0% [=====] --- B/s | 0 B ---:-- ETA
(1/13): apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64.rpm 358 kB/s | 13 kB 00:00
(2-3/13): apr-1.7.5-1.amzn2023.0.4.x86_64. 0% [=====] 354 kB/s | 13 kB 00:06 ETA
(2/13): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm 3.0 MB/s | 129 kB 00:00
(3-4/13): apr-util-1.6.3-1.amzn2023.0.2. 5% [=====] 380 kB/s | 142 kB 00:05 ETA
(3/13): apr-util-1.6.3-1.amzn2023.0.2.x86_64.rpm 2.1 MB/s | 97 kB 00:00
(4-5/13): apr-util-openssl-1.6.3-1.amzn2 9% [=====] 399 kB/s | 239 kB 00:05 ETA
(4/13): apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64.rpm 574 kB/s | 15 kB 00:00
(5-6/13): generic-logos-httpd-18.0.0-12. 10% [=====] 400 kB/s | 255 kB 00:05 ETA
(5/13): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm 747 kB/s | 19 kB 00:00
(6-7/13): httpd-2.4.68-1.amzn2023.0.1.x8 11% [=====] 404 kB/s | 273 kB 00:05 ETA
(6/13): httpd-2.4.68-1.amzn2023.0.1.x86_64.rpm 1.7 MB/s | 46 kB 00:00
(7-8/13): httpd-core-2.4.68-1.amzn2023.0 13% [=====] 413 kB/s | 319 kB 00:05 ETA
```

FIG-1: APACHE SERVER



```
[root@ip-10-0-1-152 ~]# sudo systemctl status httpd
■ httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/sbin/httpd -DFOREGROUND)
 Jul 12 08:24:46 ip-10-0-1-152.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
 Jul 12 08:24:46 ip-10-0-1-152.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
 Jul 12 08:24:46 ip-10-0-1-152.ap-south-1.compute.internal httpd[26096]: Server configured, listening on: port 80
```

FIG-2: INSTALL SERVER

➤ **STEP 5-Install PHP and MySQL Client:**

- `sudo yum install php php-mysqlnd -y`
- Verify PHP installation:
- `Php -v`

```
OpenSSH SSH client

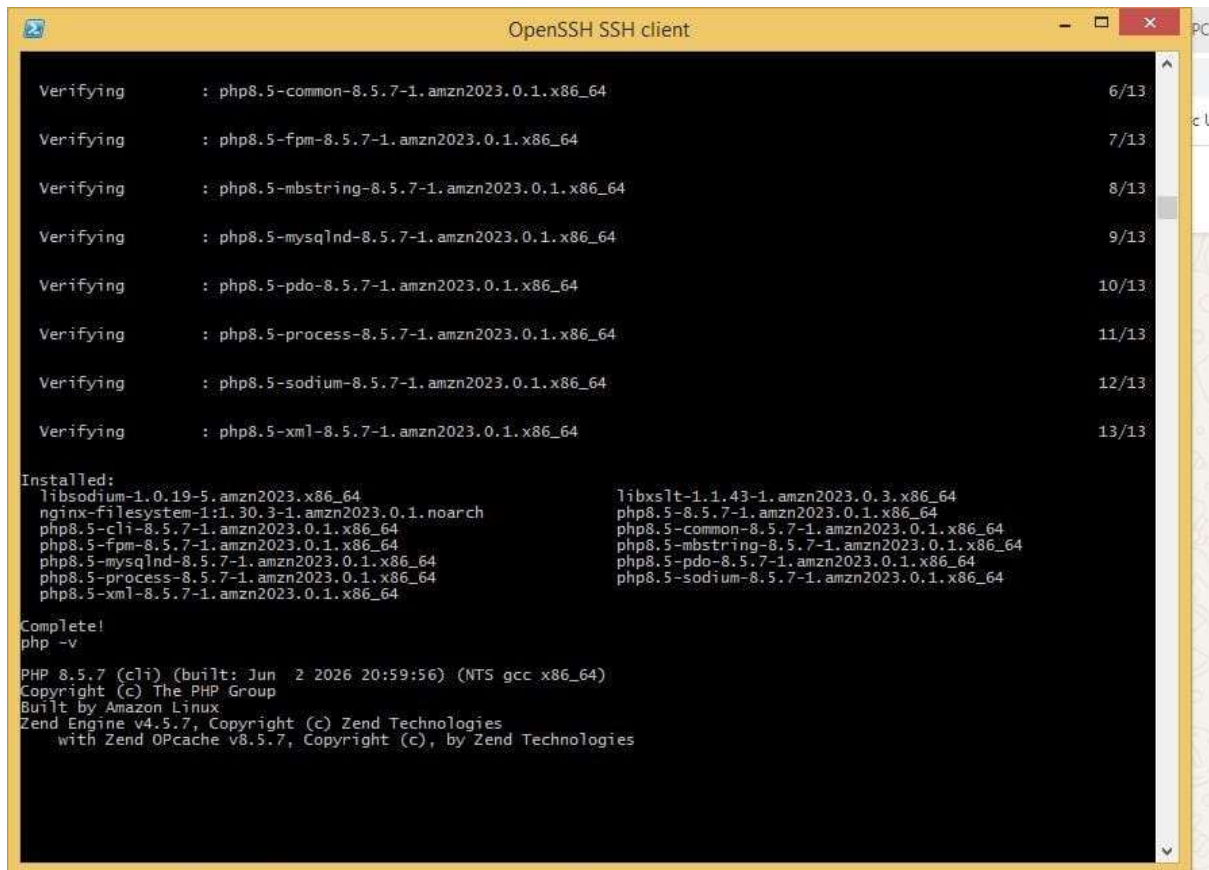
httpd.service - The Apache HTTP Server
  Loaded: loaded (/usr/lib/systemd/system/httpd.service; vendor preset: enabled)
  Active: active (running) since Jul 12 08:24:46; 1min 10s ago
  Main PID: 1234
  CGroup: /systemd/system/httpd.service
          └─ 1234 httpd -DFOREGROUND

Jul 12 08:24:46 ip-10-0-1-152.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Jul 12 08:24:46 ip-10-0-1-152.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Jul 12 08:24:46 ip-10-0-1-152.ap-south-1.compute.internal httpd[26096]: Server configured, listening on: port 80

[root@ip-10-0-1-152 ~]# sudo yum install php php-mysqlnd -y

Last metadata expiration check: 0:08:01 ago on Sun Jul 12 08:23:10 2026.
Dependencies resolved.
=====================================================================================================================================
 Package                                Architecture      Version            Repository          Size
=====================================================================================================================================
Installing:
php8.5                                 x86_64            8.5.7-1.amzn2023.0.1  amazonlinux        14 k
php8.5-mysqlnd                        x86_64            8.5.7-1.amzn2023.0.1  amazonlinux        154 k
Installing dependencies:
libsodium                             x86_64            1.0.19-5.amzn2023    amazonlinux        174 k
libxslt                               x86_64            1.1.43-1.amzn2023.0.3  amazonlinux        183 k
nginxfilesystem                       noarch            1:1.30.3-1.amzn2023.0.1  amazonlinux        10 k
php8.5-cli                           x86_64            8.5.7-1.amzn2023.0.1  amazonlinux        6.1 M
php8.5-common                         x86_64            8.5.7-1.amzn2023.0.1  amazonlinux        860 k
php8.5-pdo                           x86_64            8.5.7-1.amzn2023.0.1  amazonlinux        100 k
php8.5-process                       x86_64            8.5.7-1.amzn2023.0.1  amazonlinux        49 k
php8.5-xml                            x86_64            8.5.7-1.amzn2023.0.1  amazonlinux        248 k
Installing weak dependencies:
php8.5-fpm                           x86_64            8.5.7-1.amzn2023.0.1  amazonlinux        3.1 M
php8.5-ldap                           x86_64            8.5.7-1.amzn2023.0.1  amazonlinux        539 k
php8.5-sodium                         x86_64            8.5.7-1.amzn2023.0.1  amazonlinux        45 k
Transaction Summary
-----
Install 13 Packages
Total download size: 12 M
Installed size: 60 M
Downloading Packages:
(1/13): libsodium-1.0.19-5.amzn2023.x86_64.rpm [=====] --- B/s | 0 B --- ETA
(1/13): nginxfilesystem-1.30.3-1.amzn2023.0.1.noarch.rpm [---] --- B/s | 0 B --- ETA
(2-3/13): libsodium-1.0.19-5.amzn2023.x86_64.rpm [---] 308 kB/s | 10 kB 00:00
(2/13): libxslt-1.1.43-1.amzn2023.0.3.x86_64.rpm [---] 304 kB/s | 10 kB 00:38 ETA
(2/13): php8.5-common-8.5.7-1.amzn2023.0.1.x86_64.rpm [---] 4.0 MB/s | 183 kB 00:00
(3-4/13): php8.5-cli-8.5.7-1.amzn2023.0.1.x86_64.rpm [---] 340 kB/s | 193 kB 00:34 ETA
```

FIG-1: INSTALL PHP



```
OpenSSH SSH client

Verifying      : php8.5-common-8.5.7-1.amzn2023.0.1.x86_64      6/13
Verifying      : php8.5-fpm-8.5.7-1.amzn2023.0.1.x86_64      7/13
Verifying      : php8.5-mbstring-8.5.7-1.amzn2023.0.1.x86_64  8/13
Verifying      : php8.5-mysqlnd-8.5.7-1.amzn2023.0.1.x86_64  9/13
Verifying      : php8.5-pdo-8.5.7-1.amzn2023.0.1.x86_64      10/13
Verifying      : php8.5-process-8.5.7-1.amzn2023.0.1.x86_64  11/13
Verifying      : php8.5-sodium-8.5.7-1.amzn2023.0.1.x86_64    12/13
Verifying      : php8.5-xml-8.5.7-1.amzn2023.0.1.x86_64       13/13

Installed:
libsodium-1.0.19-5.amzn2023.x86_64
nginx-filestream-1:1.30.3-1.amzn2023.0.1.noarch
php8.5-cli-8.5.7-1.amzn2023.0.1.x86_64
php8.5-fpm-8.5.7-1.amzn2023.0.1.x86_64
php8.5-mysqlnd-8.5.7-1.amzn2023.0.1.x86_64
php8.5-process-8.5.7-1.amzn2023.0.1.x86_64
php8.5-xml-8.5.7-1.amzn2023.0.1.x86_64
libxslt-1.1.43-1.amzn2023.0.3.x86_64
php8.5-8.5.7-1.amzn2023.0.1.x86_64
php8.5-common-8.5.7-1.amzn2023.0.1.x86_64
php8.5-mbstring-8.5.7-1.amzn2023.0.1.x86_64
php8.5-pdo-8.5.7-1.amzn2023.0.1.x86_64
php8.5-sodium-8.5.7-1.amzn2023.0.1.x86_64

Complete!
php -v

PHP 8.5.7 (cli) (built: Jun  2 2026 20:59:56) (NTS gcc x86_64)
Copyright (c) The PHP Group
Built by Amazon Linux
Zend Engine v4.5.7, Copyright (c) Zend Technologies
    with Zend OPcache v8.5.7, Copyright (c), by Zend Technologies
```

FIG-2: MY SQL

➤ STEP 6-Create or Use Amazon RDS:

- Open Amazon RDS
- Create Database
- Engine: MySQL
- Template: Free Tier
- DB Identifier: studentdb
- Master Username: admin
- Password: Create a password
- Wait until the status becomes Available.

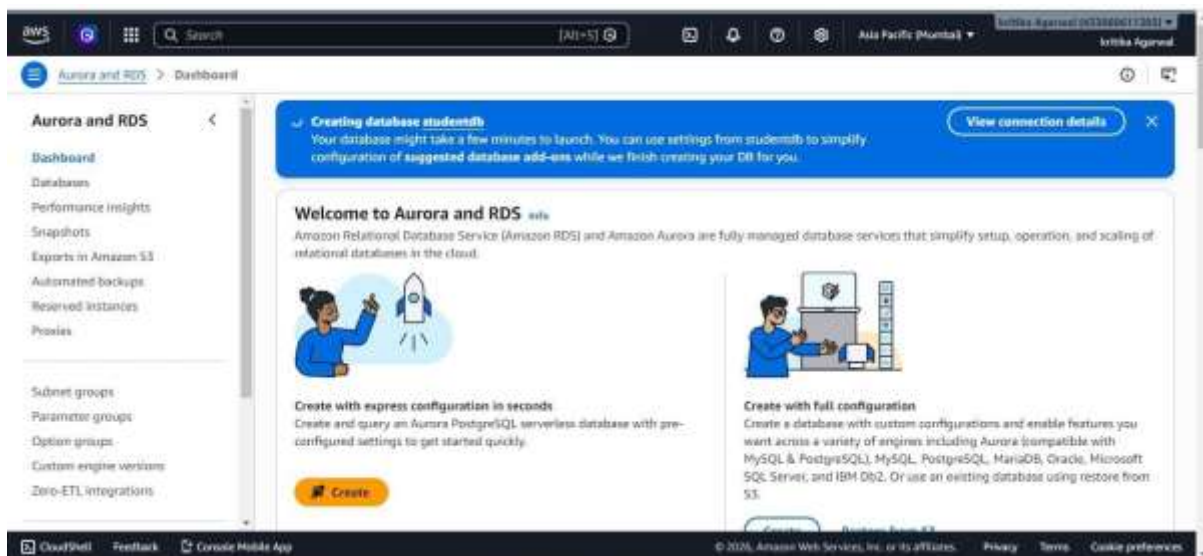


FIG-1: DB DASHBOARD

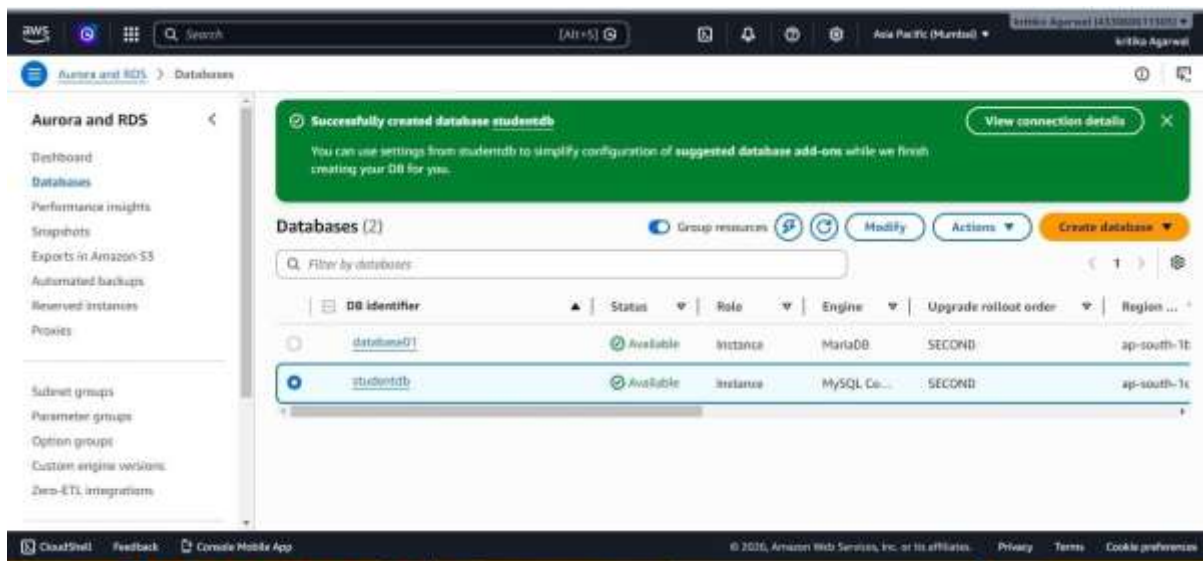


FIG-2: CREATE DB

➤ **STEP 7-Configure RDS Security Group:**

- Edit the RDS Security Group.
- Type: My SQL/Aurora
- Port: 3306
- Source: EC2 Security Group

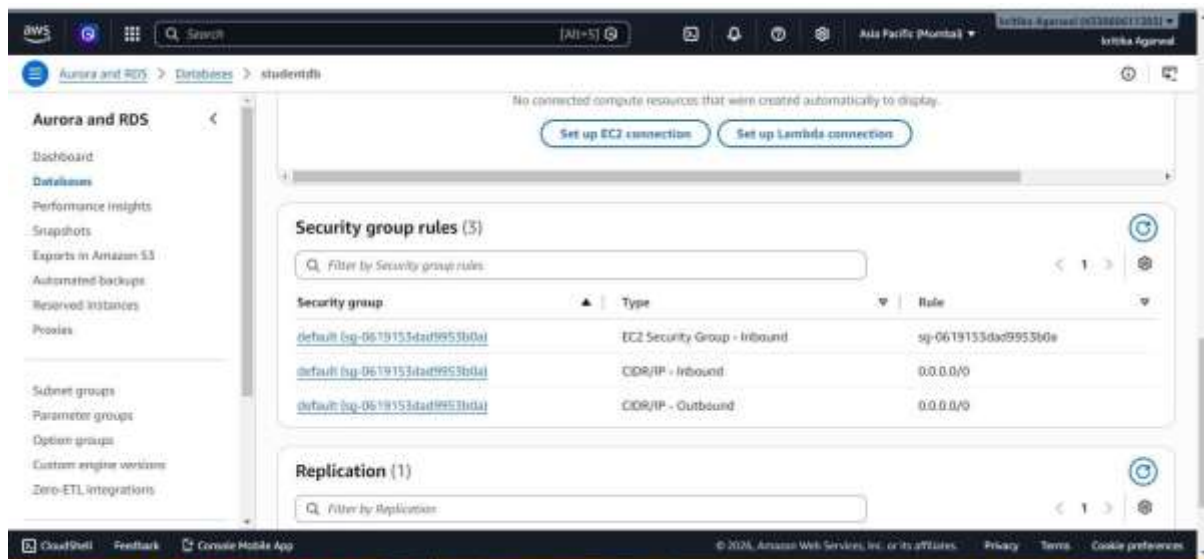


FIG-1: DATABASE SG

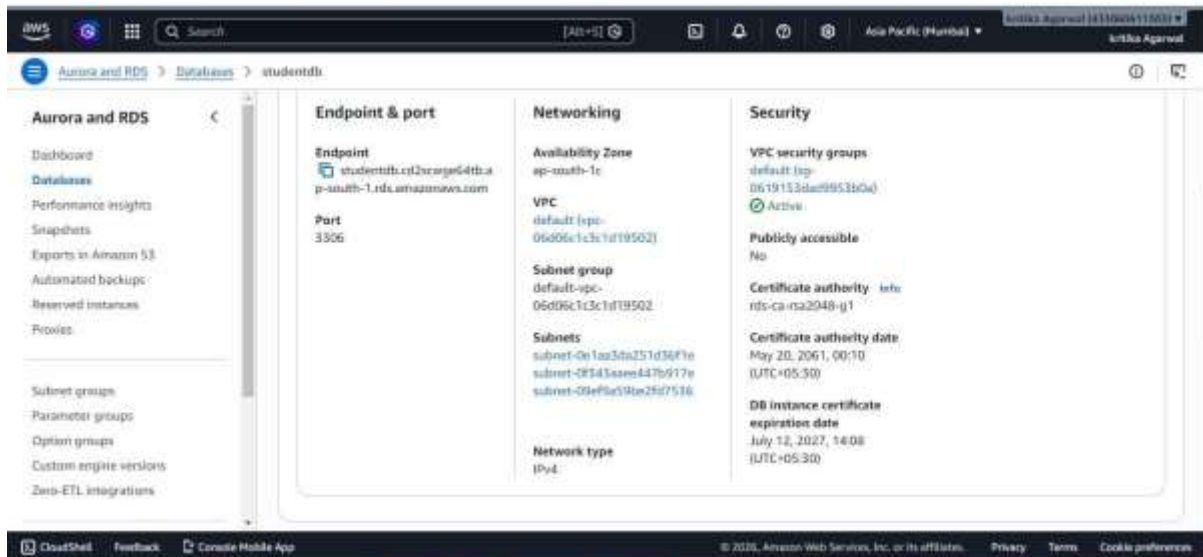
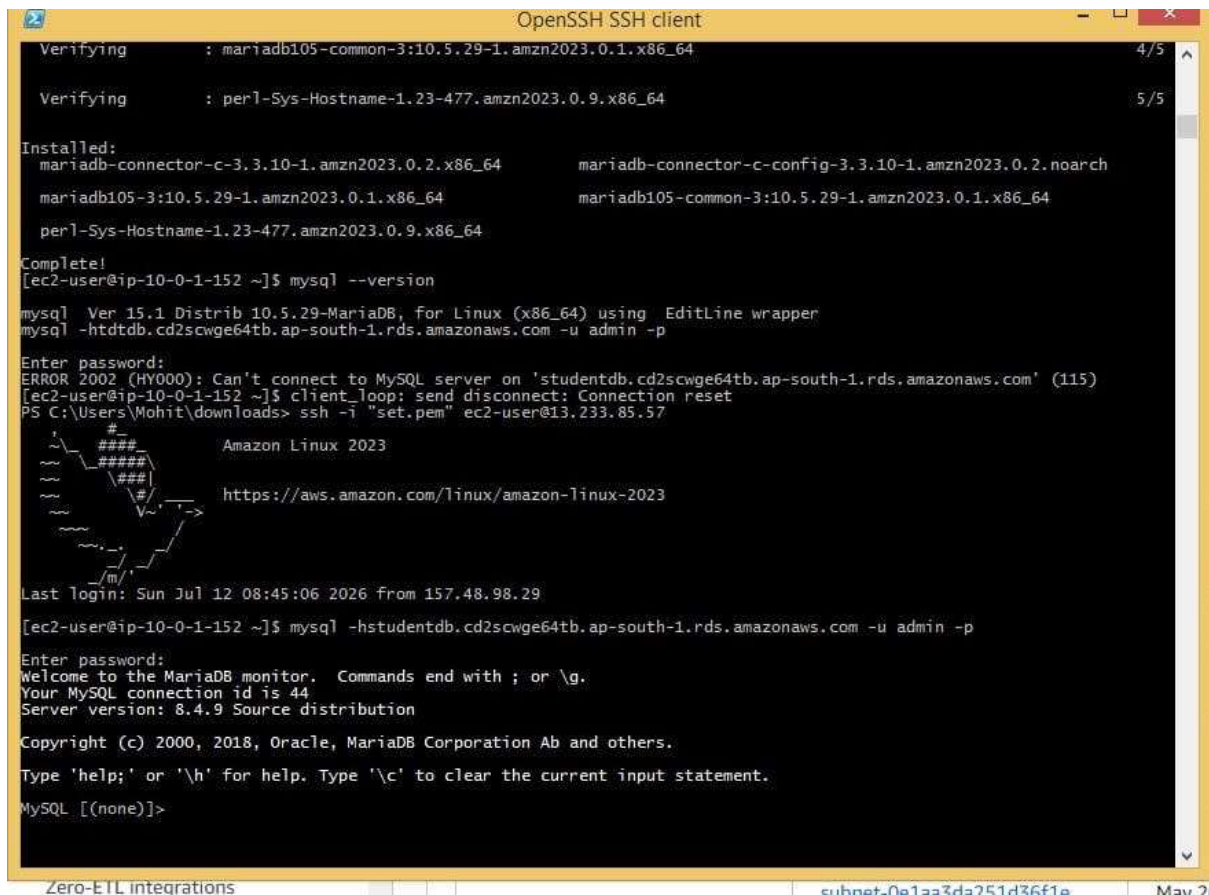


FIG-2: DB ENDPOINT

4. TESTING OUTCOMES-

➤ STEP 8-Test Database Connection:

- Install the MySQL client if needed:
- `sudo yum install mariadb105 -y`
- Connect to the database:
- `mysql -h <RDS-Endpoint> -u admin -p`
- Enter the Password.



```
OpenSSH SSH client
Verifying      : mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64 4/5
Verifying      : perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64 5/5

Installed:
mariadb-connector-c-3.3.10-1.amzn2023.0.2.x86_64      mariadb-connector-c-config-3.3.10-1.amzn2023.0.2.noarch
mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64          mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64
perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64

Complete!
[ec2-user@ip-10-0-1-152 ~]$ mysql --version
mysql Ver 15.1 Distrib 10.5.29-MariaDB, for Linux (x86_64) using EditLine wrapper
mysql -hstudentdb.cd2scwge64tb.ap-south-1.rds.amazonaws.com -u admin -p
Enter password:
ERROR 2002 (HY000): Can't connect to MySQL server on 'studentdb.cd2scwge64tb.ap-south-1.rds.amazonaws.com' (115)
[ec2-user@ip-10-0-1-152 ~]$ client_loop: send disconnect: Connection reset
PS C:\Users\Mohit\downloads> ssh -i "set.pem" ec2-user@13.233.85.57
#####
#               Amazon Linux 2023
#####
#               https://aws.amazon.com/linux/amazon-linux-2023
#####
Last login: Sun Jul 12 08:45:06 2026 from 157.48.98.29
[ec2-user@ip-10-0-1-152 ~]$ mysql -hstudentdb.cd2scwge64tb.ap-south-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 44
Server version: 8.4.9 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

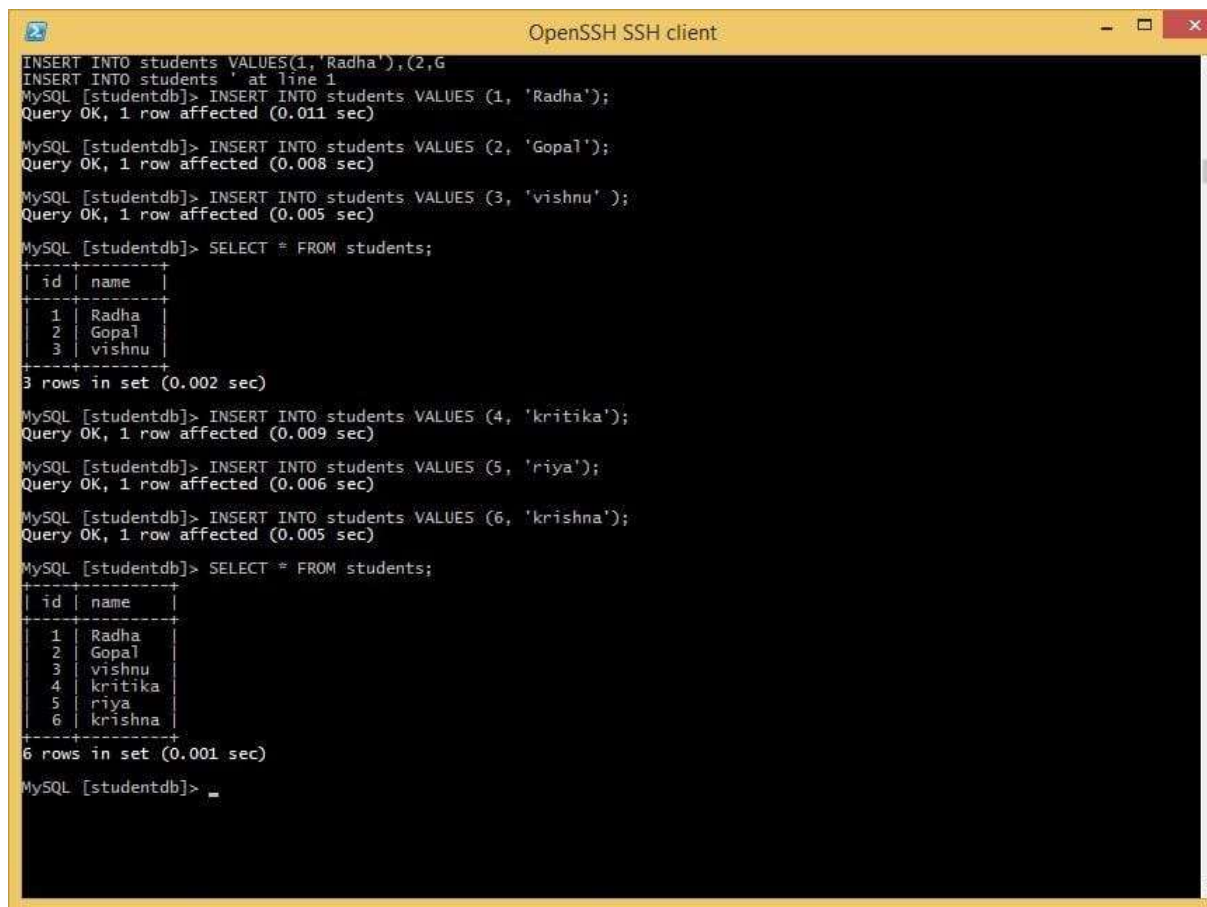
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]>
```

FIG: DB CONNECT

➤ STEP 9-Create Database and Table:

- CREATE DATABASE studentdb;
- 2. USE studentdb;
- 3. CREATE TABLE students (
 id INT PRIMARY KEY,
 name VARCHAR(50)
);
- 4. Insert data:
 INSERT INTO students VALUES
 (1,'Radha'),
 (2,'Gopal'),
 (3,'Vishnu');
- 5. Check data:
 SELECT * FROM students;



```
INSERT INTO students VALUES(1,'Radha'),(2,G
INSERT INTO students ' at line 1
MySQL [studentdb]> INSERT INTO students VALUES (1, 'Radha');
Query OK, 1 row affected (0.011 sec)

MySQL [studentdb]> INSERT INTO students VALUES (2, 'Gopal');
Query OK, 1 row affected (0.008 sec)

MySQL [studentdb]> INSERT INTO students VALUES (3, 'vishnu');
Query OK, 1 row affected (0.005 sec)

MySQL [studentdb]> SELECT * FROM students;
+----+-----+
| id | name |
+----+-----+
| 1  | Radha|
| 2  | Gopal|
| 3  | vishnu|
+----+-----+
3 rows in set (0.002 sec)

MySQL [studentdb]> INSERT INTO students VALUES (4, 'kritika');
Query OK, 1 row affected (0.009 sec)

MySQL [studentdb]> INSERT INTO students VALUES (5, 'riya');
Query OK, 1 row affected (0.006 sec)

MySQL [studentdb]> INSERT INTO students VALUES (6, 'krishna');
Query OK, 1 row affected (0.005 sec)

MySQL [studentdb]> SELECT * FROM students;
+----+-----+
| id | name |
+----+-----+
| 1  | Radha|
| 2  | Gopal|
| 3  | vishnu|
| 4  | kritika|
| 5  | riya|
| 6  | krishna|
+----+-----+
6 rows in set (0.001 sec)

MySQL [studentdb]> _
```

FIG-2:CREATE DB TABLE

➤ STEP 10-Create PHP Application:

1. Move to the web root directory:
 - `cd /var/www/html`
2. Create the PHP file:
 - `sudo nano index.php`
3. Path:
 - `/var/www/html/index.php`

5. CONCLUSION-

1. SUMMARY:

- EC2 hosts the PHP web application successfully.
- PHP connects to the Amazon RDS database.
- Student records are fetched and displayed on the webpage.
- Website is accessible using the EC2 Public IP.

2. LEARNING:

- Learned how to launch and configure an Amazon EC2 instance.
- Installed and managed Apache and PHP on Amazon Linux.
- Created and configured an Amazon RDS MySQL database.
- Connected a PHP application to Amazon RDS.
- Configured Security Groups for secure database communication.

THANK YOU
